

Effect of Stress and Glucocorticoids on the Gastrointestinal Cholinergic Enzymes

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Abstract—Male Sprague-Dawley rats maintained under controlled lighting and temperature conditions were used in this experiment. Animals were exposed to acute cold (4° C) and immobilization stress for one hour, exposed to cold stress for 7 days (chronic stress) or treated with corticosterone (2 mg/kg) 1 hr prior to sacrificing. Animals with their proper controls were sacrificed and the stomach, duodenum, ileum and colon were separated and assayed for choline acetyltransferase (ChAT) and acetylcholinesterase (AChE) activities. The data obtained indicated that exposure to acute stress resulted in significant decline in ChAT activity in all tissues studied. The administration of corticosterone resulted in significant decline in ChAT activity in all tissues studied except for the duodenum. Meanwhile, the exposure to chronic stress did not have any significant effect on ChAT activity. On the other hand, acute stress caused significant increase in AChE activity in all tissues studied except for the ileum and stomach. The duodenal AChE activity of stressed animals increased thirty-fold when compared to control. The administration of glucocorticoids significantly reduced AChE in all tissues studied, except for the duodenum and stomach where there was thirty-two-fold increase as compared to the control levels. The exposure to chronic stress also caused significant increase in AChE of all tissues studied, except for the colon. The results of this experiments indicate that the duodenal AChE is extremely sensitive to stress or glucocorticoids and that stress induced changes in the cholinergic enzymes of the gastrointestinal tract may be mediated by adrenal steroids.

Introduction

Cholinergic activity exerts a considerable influence on the gastrointestinal function (Bunce *et al.*, 1977). These cholinergic influences are controlled by long vago-vagal reflexes (Debas *et al.*, 1975) and by local cholinergic reflexes which have been identified in the absence of vagal innervation (Magee and Hu, 1975). Ulcer formation has been linked to acid secretion (Anderson, 1967). Gastric emptying and acid secretion were found to be controlled by cholinergic receptors in the duodenum (Hunt and Knox, 1968). A possible role for a local cholinergic mechanism in the control of acid secretion has been studied using atropine in animals in which the vagal innervation to stomach or its fundic pouch has been severed (Hunt and Knox, 1968). Vizi *et al.* (1972, 1973) have demonstrated that gastrin can release acetylcholine from the myenteric plexus in the guinea pig ileum. They also suggest that the secretagogue action of gastrin in the stomach is partly mediated through the release of acetylcholine.

Recent results from this laboratory have shown that there is an inhibition in AChE activity in rat hypothalamus and the adrenal gland when animals were exposed to acute or chronic stress or after the administration of corticosterone (Gabriel and Soliman, 1983). These effects might be due to the decreased synthesis and/or increased degradation of the enzyme. It was also observed that the administration of corticosterone to intact animals caused the same effect on AChE as in the case of stress exposure which indicates that stress-induced decline of AChE might be mediated through the adrenal cortex (Gabriel and Soliman, 1983).

For many years, investigators have examined the pathogenesis of stress-induced ulcers. In spite of numerous studies and hypothesis, it appears that no single etiological mechanism can be pinpointed as the causative factor for the development of these ulcers. At present these ulcers are viewed as a multifactorial phenomenon. One of the factors studied extensively has been the role of the adrenal gland. Weiss (1971) had demonstrated that rats that showed very high steroid levels also showed very severe ulcerations. Meanwhile, Murphy *et al.* (1979) showed that in stressful situations, animals with hippocampal lesions have enhanced adrenocortical activity with an increase in the incidence of gastric ulcers. On the other hand, stress and glucocorticoids have been implicated in different gastrointestinal diseases. Emotional disturbance and physical stress have been linked to diarrhea (Sleisenger and Fordtran, 1978) and to irritable bowel syndrome (Almy, 1976).

Therefore, the present investigation was designed to study the effect of stress or glucocorticoid on the acetylcholine synthesizing enzyme (choline acetyltransferase, ChAT) and acetylcholinesterase, (AChE) in different regions of the gastrointestinal tract.

Methods

Male Sprague-Dawley rats weighing 130–160 g and purchased from Southern Animals Farms (Prattville, AL) were used in this study. They were adapted to a temperature of $21 \pm 1^\circ\text{C}$ in an environmental chamber equipped to provide 23.25 cd/m^2 of cool, white fluorescent light for a minimum period of one week. The light period in the chamber was automatically timed to last from 08.00 to 20.00 hr followed by a 12-hour dark period. Standard pellet diet (Purina, St. Louis, MO) and water were provided ad libitum.

In this experiment, 24 rats were randomly divided into 4 groups. In the first group, animals were exposed to cold stress at 4°C of ambient temperature; they were also simultaneously immobilized for 1 hr (acute stress group). The second group was subjected to the same cold stress (without immobilization) for 7 days (chronic stress group). Another group of animals were injected with corticosterone (Sigma Chemical Company St. Louis, MO; 2 mg/kg, i.p.) 1 hr prior to sacrificing. Six other animals were kept under normal environmental conditions and were used as a control group.

All the animals were sacrificed by decapitation at 10.00 hr. The stomach, duodenum, ileum and colon were rapidly separated on ice and the mucosa of each region was gently scrapped by means of a scalpel. AChE was assayed using the procedure of Ellman *et al.* (1961). Each tissue was individually homogenized (1 % W/V) in 0.1 M ice-cold phosphate buffer (pH 7.4) containing 0.1 % Triton X-100 (Sigma). AChE activity in each homogenate was immediately determined by the spectrophotometric method and expressed as micromoles of substrate (acetylthiocholine iodide; Sigma) hydrolyzed per minute per gram of tissue.

For ChAT activity each tissue was individually homogenized (1 % W/V) in 0.1 M ice-cold phosphate buffer (pH 7.0). ChAT activity was determined using the spectrophotometric method of Chao and Wolfgram (1972). ChAT activity was expressed as nanomoles of COASH formed per minute per gram of tissue.

All data were analyzed using a Hewlett-Packard 9815A Computer. Overall comparisons between the different groups were performed by one-way analysis of variance (Steel and Torrie, 1960). The significance of the difference between the means of various tissues was determined using the LSD test (Steel and Torrie, 1960).

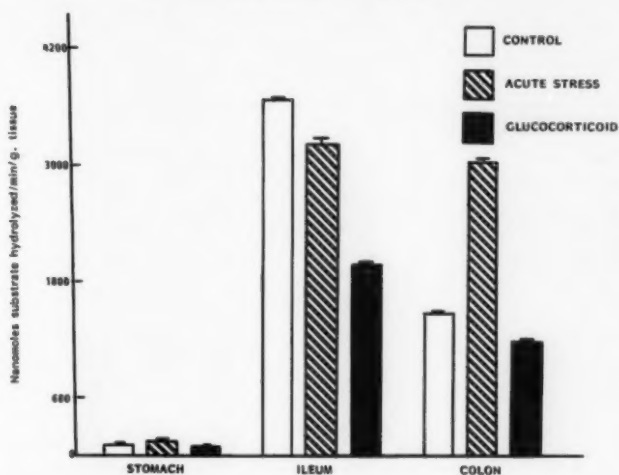


FIG. 1

Effect of immobilization and cold stress (acute stress) or glucocorticoid (2 mg/kg) treatment on the activity of AChE in the stomach, ileum and colon. Each bar represents the mean of 6 animals $\pm$ SEM.

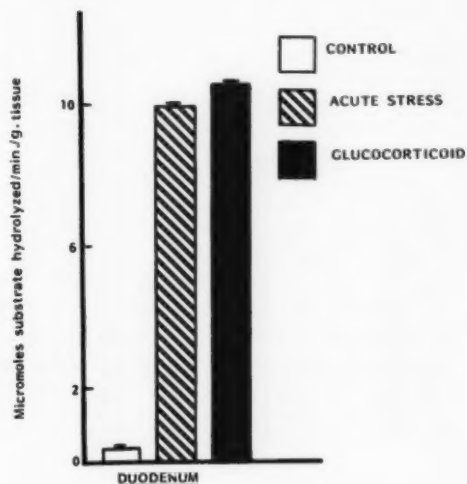


FIG. 2

Effect of immobilization and cold stress (acute stress) or glucocorticoid (2 mg/kg) treatment on the activity of AChE in the duodenum. Each bar represents the mean of 6 animals $\pm$ SEM.

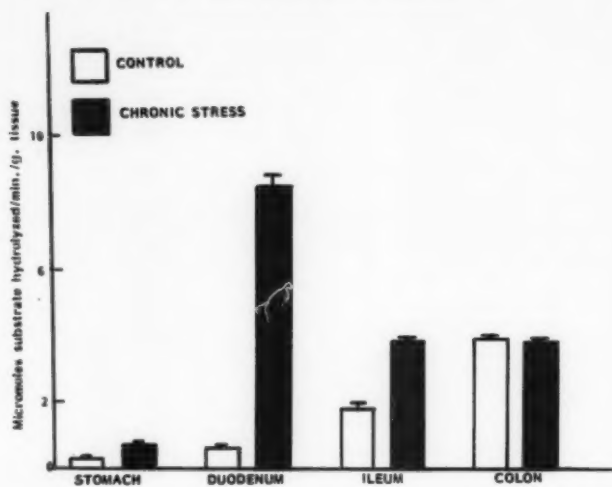


FIG. 3

Effect of cold exposure for 7 days at 4°C (chronic stress) on the activity of AChE in the stomach, duodenum, ileum and colon. Each bar represents the mean of 6 animals $\pm$ SEM.

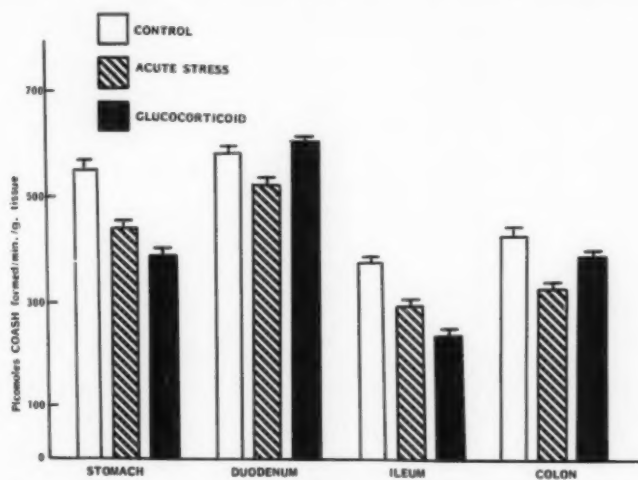


FIG. 4

Effect of immobilization and cold stress (acute stress) or glucocorticoid (2 mg/kg) treatment on the activity of ChAT in the stomach, duodenum, ileum and colon. Each bar represents the mean of 6 animals $\pm$ SEM.

Results

Figures 1 and 2 show the effect of acute stress or glucocorticoid treatment on the AChE activity in different regions of the gastrointestinal tract. In comparison with untreated controls, acute stress caused a significant increase ($p < 0.001$) in enzyme activity in the duodenum (3145 %), colon (75 %), and a significant decrease ($p < 0.001$) in enzyme activity in the duodenum (3334 %) and a significant decrease ($p < 0.01$) in enzyme activity in the ileum (44.3 %), and colon (33.0 %) as shown in Figs. 1 and 2. There was no significant change in AChE activity in the stomach ($p > 0.05$).

Figure 3 depicts the activity of AChE in chronic stress animals compared to the control group. The results show significant increase ($p < 0.05$) in enzyme activity in the stomach (54.5 %), duodenum (1627 %), ileum (121 %) and no significant change in the colon.

Figure 4 represents the activity of ChAT in the different gastrointestinal tract regions studied in control, acute stress rats and corticosterone treated animals (2 mg/kg i.p.). The results show that there was significant decline ($p < 0.05$) in the stomach (19.1 %), duodenum (10.3 %), colon (24.7 %), and ileum (21.0 %) of the acute stress groups. In comparison with the control group, corticosterone treatment resulted in a significant decrease ($p < 0.05$) of ChAT activity in the stomach (29.1 %), colon (9.41 %) and ileum (35.1 %) with no significant effect in the duodenum.

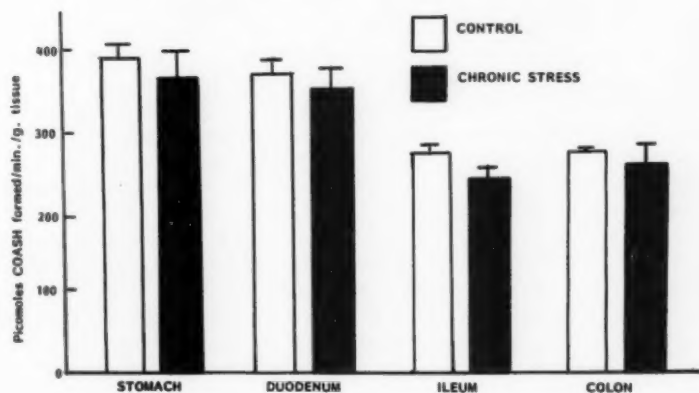


FIG. 5

Effect of cold exposure for 7 days at 4° C (chronic stress) on the activity of ChAT in the stomach, duodenum, ileum and colon. Each bar represents the mean of 6 animals $\pm$ SEM.

Figure 5 summarized ChAT activity in the different regions studied in control and chronic stress animals. There was no significant change in ChAT activity in all tissues studied.

Discussion

The results of the study indicate that the activity of AChE and ChAT varies in different regions of the gastrointestinal tract, and that the duodenal AChE is extremely sensitive to stress or glucocorticoid administration. The current findings also suggest that stress-induced changes in the cholinergic enzymes of the gastrointestinal tract may be mediated by adrenal steroids.

The fact that administration of corticosterone to intact rats caused the same effect as that seen in acute stress animals, gives support to the fact that stress-induced changes in AChE may be mediated by adrenal steroids. Although we have not measured the levels of corticosterone in the plasma, it has been reported that following acute or chronic exposure to cold stress, a rise in the circulating levels of corticosterone was noted in the rat (Sakellaris and Vernikos-Danellis, 1975). Tache' *et al.* (1976) noted that female rats exposed for 3 days of stress showed an inhibition in body weight and an induced adrenal enlargement. These authors also showed that after six days of stress there was stabilization in the adrenal weight. An increase was also noted in the plasma levels following 150 minutes of daily immobilization (Mikulaj *et al.*, 1974).

Incidence of gastric and peptic ulcers have been related to stress (Murphy *et al.*, 1979). Some probable causes are hypersecretion of gastric acid and the development of ischemic areas in the gastroduodenal mucosa (Leonard *et al.*, 1964). Leonard *et al.* (1964) have postulated that any severe stress may cause prolonged ischemia of the gastroduodenal mucosa. They note that strong sympathetic stimulus and high circulating catecholamines caused by severe stress may open arteriovenous shunts in the gastroduodenal submucosa. Because of this phenomenon, blood which normally flows through the capillary bed of the gastroduodenal mucosa, then flows instead through the submucosal arteriovenous shunts and is thus diverted from the gastroduodenal mucosa. As a result of this, the mucosa is severely injured. The ulceration develops when areas of mucosa undergo necrosis. Anticholinergic drugs have been used to treat these ulcers on the assumption that they shunt blood away from the bleeding mucosa by blocking excessive splanchnic parasympathetic activity and thus opening submucosal arteriovenous shunts (Leonard *et al.*, 1964).

Choline acetyltransferase activity has frequently been used as a reliable marker of the cholinergic nerve activity. In this study, biochemical measurements of ChAT and AChE following stress revealed changes in

cholinergic activity of the mucosa of gastrointestinal tract region studied. Similar results were obtained when animals were treated with corticosterone. The results obtained are an indication of the mucosal reaction when the body is exposed to stress. The decline in the cholinergic function of different regions of the gastrointestinal tract may explain the ischemia produced in gastroduodenal mucosa following stress (Leonard *et al.*, 1964). The results also indicate that adaptation can take place in the cholinergic activity when animals were exposed to chronic stress conditions. There was no observed change in ChAT activity in animals exposed for 7 days to cold stress.

It is interesting to note the rapidity of the activation of AChE in the duodenum. The dramatic increase in AChE is probably associated with a direct activating effect on the enzyme rather than an increase in the rate of enzyme synthesis in the gastrointestinal mucosa.

Although the outcome of this investigation was not expected, the present work provides very significant information regarding the role of stress and glucocorticoids in the regulation of the cholinergic enzymes of the gastrointestinal tract. Additional information could be obtained by assaying for the actual levels of acetylcholine in the gastrointestinal tract mucosa in relation to stress.

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